

Question:

Calculate the quantity of heat required to (a) convert a liter of H<sub>2</sub>O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

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1 . To calculate the heat required to convert a liter of water at 30 °C to a liter of water at 60 °C , we need to calculate the heat required for the phase change ( from solid to liquid and then from liquid to gas , but since we are considering it as liquid , we only consider the liquid to liquid transition ) and the heat required to raise the temperature of the water from 30 °C to 60 °C .

- The specific heat of water is 1 cal /g °C .
- A liter of water is approximately 1 000 grams .
- The change in temperature ( ΔT ) is 60 °C - 30 °C = 30 °C .
- The heat required for temperature change ( Q<sub>temp</sub> ) is given by the formula  $Q_{temp} = mass \times specific\ heat \times \Delta T$  , which is  $1\,000 \times 1 \times 30 = 30\,000$  calories .
- For the phase change ( free zing and melting ), the heat required will be the latent heat of fusion , which for water is about 80 cal /g .
- Since the process can be simplified by assuming the water is initially in liquid state and directly heated , we use the average specific heat ( 1 cal /g °C ) to estimate the heat for this part , as the latent heat would be negligible in a small temperature change .

2 . The heat required to heat a 1 kg ( 1 000 grams ) block of aluminum from 30 °C to 60 °C can be calculated using the formula  $Q = mass \times specific\ heat \times \Delta T$  .

- The specific heat of aluminum is 0 . 2 1 5 cal /g °C .
- ΔT is 30 °C .
- Therefore ,  $Q_{aluminum} = 1\,000 \times 0 . 2\,1\,5 \times 30 = 6\,450$  calories .

3 . Add the two quantities of heat :

- Total heat ( Q ) =  $Q_{temp} + Q_{aluminum} = 30\,000 + 6\,450 = 36\,450$  calories .
- Convert the total heat to Kilo-cal ories ( since 1 Kcal = 1 000 calories ):  $Q_{total} = 36\,450 / 1\,000 = 36 . 45$  Kcal .
- Since the phase change heat is negligible , the slight difference in the options can be due to rounding or approximation error , but the most accurate answer in the options is 30 Kcal ( for the temperature change part ) and 6 . 3 Kcal ( for the aluminum heat part ).

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A . 30 Kcal , 6 . 3 Kcal

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